AMITY UNIVERSITY
PATNA

Course Curriculum

Course Title: Quantitative Techniques
Course Code : QAM303
Credit Units: 02
Course Level : UG
Syllabus Version-I

L	T	P/S	SW	No. of PSDA	Total Credit Unit
2	0	0	0	0	2

Course Description:

This course provides a comprehensive foundation in quantitative aptitude, covering essential mathematical concepts and their real-world applications. Beginning with the number system, HCF, LCM, and simplification techniques, students develop computational efficiency. Applied arithmetic introduces logarithms, percentages, and algebraic foundations for analytical reasoning. Business and speed mathematics focus on profit and loss, ratio and proportion, and practical problem-solving in areas like time, distance, and work-related calculations. Advanced quantitative concepts extend to financial mathematics, permutations, combinations, and basic probability, equipping students with the skills needed for competitive exams and professional problem-solving in various domains.

Course Objective:

CO1: Remember: Define and recall fundamental concepts of the number system, HCF, LCM, simplification techniques, and core principles of applied arithmetic, business mathematics, and quantitative aptitude.
CO2: Understand: Explain the properties of arithmetic operations, algebraic foundations, probability, and financial mathematics, demonstrating their relevance in real-world applications.
CO3: Apply: Solve numerical problems related to HCF, LCM, logarithms, percentages, profit and loss, time-distance calculations, permutations, and probability using appropriate mathematical techniques.
CO4: Analyze: Differentiate between various quantitative problem-solving strategies, assessing their effectiveness in competitive exams and professional contexts.

Pre-requisites: Basic knowledge of Mathematics.

Contents/Syllabus:

	Weightage (%)	Lecture Hours
Module I: Number System	**25**	**3**
Number system, HCF and LCM of numbers, Decimal fractions, Simplification, Square roots and cube roots, Problems on numbers.		
Module II: Applied Arithmetic and Algebraic Foundations	**25**	**4**
Average, Problems on ages, Surds and indices, Logarithms, Percentage, Chain rule		
Module III: Business and Speed Mathematics	**25**	**4**
Profit and loss, Ratio and proportion, Partnership problems, Pipes and cisterns, Time and work, Time and distance, Boats and Streams, Problems on Trains.		
Module IV: Advanced Quantitative Concepts	**25**	**4**
Simple interest, Compound interest, Area, Volume and surface area, Calander, Clocks, Permutations and combinations, Probability (basic).		

Course Learning Outcomes (CLOs):

CLO1: Demonstrate knowledge of number systems, HCF and LCM, percentages, logarithms and simple and compound interest.
CLO2: Explain problems related to averages and ages using logical reasoning
CLO3: Describe profit and loss scenarios and apply ratio and proportion techniques in financial contexts.
CLO4: Apply permutations, combinations, and basic probability principles for logical decision-making.

Pedagogy for Course Delivery:

The course will be taught in theory-based mode. The instructor will discuss quantitative technique problems to the students for better understanding of theconcept.

Assessment/ Examination Scheme:

Theory L/T (%)	Lab/Practical/Studio (%)	TOTAL
60	40	100

Theory Assessment (L&T):

Continuous Assessment/Internal Assessment (40 %)					End Term Examination (60%)
Components (Drop down)	Viva	Class Test	Home Assignment	Attendance	End Term
Weightage (%)	5	20	10	5	60

SN	Bloom's Level	Course Learning Outcomes	Class Test	Viva
1	**Understand**	Understand the properties of arithmetic operations, algebraic foundations, probability, and financial mathematics, demonstrating their relevance in real-world applications.	✓	✓
2	**Apply**	Apply knowledge to solve numerical problems related to HCF, LCM, logarithms, percentages, profit and loss, time-distance calculations, permutations, and probability using appropriate mathematical techniques.	✓	✓
3	**Analyze**	Analyze various quantitative problem-solving strategies, assessing their effectiveness in competitive exams and professional contexts.	✓	✓
4	**Evaluate**	Critically evaluate mathematical models for financial calculations, ratio analysis, and business problem-solving, ensuring accuracy and optimal decision-making.	✓	✓

Text:

1. R.S. Aggarwal, Quantitative Aptitude, S. Chand & Company, New Delhi, 2012.
2. Govind Prasad Singh and Rakesh Kumar, Text Book of Quickest Mathematics (for all Competitive Examinations), Kiran Prakashan, 2012.
3. R.S. Aggarwal, Objective Arithmetic, S. Chand & Company, New Delhi, 2005.

References:

- "UGC NET/SLET: Teaching & Research Aptitude", Bright Publications, 2010.
- Dr. Lal, Jain, Dr. K. C. Vashistha, "U.G.C.- NET/JRF/SET Teaching & Research Aptitude", Upkar Prakashan, 2010.

Web Resources

- https://www.studocu.com/in/document/bangalore-university/bachelor-of-computer-applications/quantitative-techniques/81587629
- https://www.geektonight.com/quantitative-techniques-and-methods-notes